
 $^{40}\text{Ca}(^3\text{He}, ^8\text{Li})$ [1976Be08](#)

[1976Be08](#): ^{35}K isotope discovery. $^{40}\text{Ca}(^3\text{He}, ^8\text{Li})^{35}\text{K}$ is studied using 73.7- and 75.8-MeV ^3He beams produced by the Michigan State University cyclotron. Targets were 370 and 190 $\mu\text{g}/\text{cm}^2$ enriched ^{40}Ca on a 20 $\mu\text{g}/\text{cm}^2$ natural carbon backing. ^8Li particles were detected using a scintillator-proportional counter detector system at the focal plane of the Enge split-pole spectrograph. Measured the cross section for producing ^{35}K g.s. and the mass excess of ^{35}K . Also see [1976BeXJ](#) and [1976BeZJ](#).

 ^{35}K Levels

<u>E(level)</u>	<u>Comments</u>
0	
1560 40	Assigned by 1976Be08 as the mirror state of $1/2^+$ level almost at the same energy in ^{35}S .
2690 50	Assigned by 1976Be08 as the mirror state of $5/2^+$ level almost at the same energy in ^{35}S .